



INNOVATING CLEAN SPACES

Grade **04**

ALIGNED WITH INTERNATIONAL STANDARDS

This book supports essential educational frameworks that shape future-ready minds:

- 21st Century Skills
- Social-Emotional Learning (SEL)
- Sustainable Development Goals (SDGs)
- Universal Design for Learning (UDL)

STEAM Scouts is a ground-breaking STEAM book series with an interdisciplinary focus designed to ignite curiosity, creativity, and critical thinking in learners worldwide. The series supports Social-Emotional Learning (SEL) by promoting core competencies like self-awareness, self-management, social awareness, relationship skills, and responsible decision-making. It also aligns with the UN Sustainable Development Goals (SDGs), fosters 21st Century Skills, and meets international educational standards to equip learners with tools for a dynamic, innovative, and impact-driven future. Furthermore, it is built on Universal Design for Learning (UDL) principles, ensuring inclusive, accessible, and engaging STEAM education for all students, regardless of learning ability.

Key Features

Standards-Aligned

- NGSS (Next Generation Science Standards) – Authentic scientific inquiry & engineering design.
- CCSS (Common Core State Standards) – Rigorous literacy & math integration.
- Cambridge & IB (PYP/MYP/DP) – Global perspectives & interdisciplinary learning.
- ISTE (Technology Standards) – Digital literacy & computational thinking.
- STEL (Standards for Technological & Engineering Literacy) – Real-world problem-solving.

Social-Emotional Learning (SEL) and Global Citizenship

- Each book fosters collaboration, resilience, leadership, and ethical decision-making while connecting to multiple UN Sustainable Development Goals (SDGs) - such as Quality Education (SDG 4), Clean Energy (SDG 7), Climate Action (SDG 13), and Innovation (SDG 9) - empowering students to think globally, encourages ethical innovation & community impact.
- Differentiated challenges to meet diverse learning needs and skill levels.
- Visual cues and infographics to enhance understanding and engagement.
- Guided instruction to support learners through new or complex concepts.
- Scaffolded activities that build skills progressively and promote learner independence.

Why Choose This Series?

Future-Ready Skills – Prepares learners with essential skills like creativity, collaboration, and critical thinking for future careers and global challenges.

Teacher & Student-Friendly – Includes engaging lesson hooks, discussion prompts, guided instruction, and hands-on tasks for seamless learning.

- **SDG-Themed Modules** – Each module links to 2+ UN SDGs, blending academics with global purpose and ethical action.
- **Inclusive & Engaging** – Built on UDL principles to ensure accessibility through differentiated pathways, visual supports, and hands-on learning.

Perfect For:

- K-12 Schools adopting IB, Cambridge, NGSS, or CCSS
- STEM/STEAM Coordinators seeking industry-relevant curricula
- Homeschoolers and Educators wanting SDG-aligned, future-focused content
- Educational Institutions committed to sustainability and inclusion

Theme

03

INNOVATING CLEAN SPACES

Each week has a double session, total time of session 80-90 mins.



Week	Unit Name	Lesson	Page
Week 1	Unit 1: Spark The Challenge!	Lesson 1: Don't Trash It, Transform It!	2-24
Week 2		Lesson 2: Innovate to Elevate.	25-38
Week 3		Lesson 3: Building Our Green Future.	39-62
Week 4	Unit 2: Innovators in Action.	Lesson 1: Power Up Your ideas!	63-81
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Week 7	Unit 3: Beyond Imagination.	Lesson 1: Smart Sensors in Action.	124-150
Week 8		Lesson 2: Speak Up, Smart Bin!	151-173

About The Theme

Everyone deserves access to clean, healthy, and welcoming public spaces—but many schoolyards and communities still struggle with overflowing bins, unsorted waste, and poor recycling habits. In this theme, students will become innovative engineers and environmental technologists, designing a Smart Litter Basket that uses coding, electronics, and artificial intelligence to help keep shared spaces cleaner, smarter, and more sustainable.

As students explore the real-world challenges of waste management, they'll investigate the reasons why people avoid using trash bins and how design can influence better behavior. They'll apply their knowledge of sensors and automation—using components like moisture sensors, IR motion detectors, and servo motors—to create hands-free bins that sort waste, detect fullness, and promote hygienic use.

Through hands-on experimentation, students will dive into the foundations of electronics and coding. They'll explore Ohm's Law, build circuits, and write programs using conditional logic, loops, and sensor feedback. Using tools like Arduino and Tinkercad, they'll develop fully functional smart bins that integrate LEDs, LCD displays, and other interactive components. Students will also explore how renewable energy sources—like solar panels—can power their devices sustainably.

As a forward-thinking extension, students can explore how AI can be used to improve waste sorting, recognize patterns in disposal behavior, and adapt bin functionality to meet the needs of different users—pushing their projects into the realm of intelligent systems.

This project supports Sustainable Development Goals 11 (Sustainable Cities and Communities) and 12 (Responsible Consumption and Production), encouraging students to combine innovation, empathy, and environmental responsibility. By the end of the theme, they won't just understand recycling—they'll have built a smart, tech-powered solution that helps everyone live in cleaner, more inclusive, and more sustainable communities.

HOW TO USE THIS BOOK?

1

Ready, Set, Learn!

Where we merge your existing knowledge with exciting new discoveries.

2

Inquiry-Based Learning!

Let's Think!

In this section, students will brainstorm and explore possible solutions to specific problems. This encourages critical thinking and creativity as they consider various approaches.

Explorer's Toolkit

Here, students will dive deeper into the topic. They will engage with curated videos, articles, educational games, and hands-on experiments to broaden their understanding and spark curiosity.

DIY Lab

It's part of the explore, students will undertake hands-on projects where they will design, build, and test their own creations. They will apply scientific principles, use technology and engineering skills, explore artistic concepts, and utilize mathematical reasoning.

Screen Lessons

Students will watch videos that explain key concepts in an engaging and accessible way.

Dig Deeper

Students will read articles that provide deeper insights into specific concepts, enhancing their understanding.

3

Problem-Based Learning:

- **Introducing the problem:**
Students will join a journey to tackle real-world environmental challenges. Through hands-on activities and engaging tasks, they will apply their knowledge and skills to discover innovative solutions.
- **Apply the Engineering Design Process (EDP):**
Students will use a structured approach to define problems, conduct research, generate ideas, prototype solutions, test, evaluate, and refine their designs.
- **The Scientific Method:**
Students will follow a systematic process of inquiry—making observations, forming hypotheses, conducting experiments, analyzing data, and drawing conclusions—to develop a deeper understanding of scientific concepts and problem-solving. In doing so, they will incorporate the CER framework to present their scientific claims, provide evidence from their experiments, and explain their reasoning behind the conclusions they draw.

B

4

Project-Based Learning:

PBL immerses students in real-world challenges through hands-on projects. They build critical thinking, creativity, collaboration, and communication skills while designing solutions and taking ownership of their learning.

5

Story

Join Adam and Laila, two fearless STEAM Scouts, on adventures full of discovery and creativity! From building inventions to solving real-world challenges, they'll inspire you to think, experiment, and innovate. Step into the world of science, technology, engineering, arts, and math—where every challenge becomes an adventure!

Story Reflection

In this section, students will reflect on the story's key themes, characters, and challenges.

6

Assessment Exercises:

These are divided into three categories to ensure comprehensive learning and mastery.

Apprentice:

Developing understanding and can apply skills with guidance.

Builder:

Demonstrates solid understanding and can apply skills independently.

Mastermind:

Demonstrates deep understanding and can apply skills creatively and strategically.

7

Career Readiness

This section equips students with essential skills for future careers by fostering problem-solving, teamwork, and adaptability. Through hands-on projects, real-world applications, and critical thinking challenges, students develop the confidence and competencies needed for success in the ever-evolving job market.

8

Mission Innovate

Now, it's your turn! In Mission Innovate; students will apply their creativity, problem-solving skills, and knowledge to develop their own competition projects. This challenge encourages independent thinking, collaboration, and real-world innovation, empowering students to become the next generation of creators and problem-solvers.

Now I Can

This section is designed to help you measure your progress and achievements. Enjoy your exploration and learning journey through this STEAM book!

My Journal

In this part students will write their own story while they are creating the project.

By the end of this theme, students will be able to:

- Explore the impact of improper waste disposal on ecosystems and communities
- Understand the importance of recycling, waste sorting, and smart waste management in reducing pollution.
- Compare recyclable and non-recyclable materials based on their properties and impact on the environment.
- Use an AI-powered mapping tool to generate and analyze a world map of recycling bin color codes, comparing similarities and differences across countries.
- Identify and compare physical and chemical changes to explain which materials can be reused and which cannot.
- Explore how parallel circuits work by understanding how electric current splits and how voltage and current behave across multiple paths.
- Build simple circuits using a breadboard and Arduino, and understand how to connect components like LEDs and motors.
- Understand how an IR sensor detects motion and how it can be used to trigger actions in a circuit.
- Integrate an IR sensor with a servo motor to automate movement, such as opening a bin lid when motion is detected.
- Understand the importance of separating wet and dry waste for effective recycling, and explore how a moisture sensor can help detect the difference.
- Recognize the difference between analog and digital components, and apply planning, measuring, and safety skills when building a project.
- Design and build a smart trash bin that uses coding and hardware to sense when it's full and respond automatically, while developing real-world problem-solving skills.
- Understand how sensory signals like sound and light can be used to communicate important information in smart devices.
- Explore how to build eco-friendly systems by using Arduino with solar panels and LCDs to display helpful messages.

Unit

1

Lesson

1

Spark The Challenge!

Don't Trash It, Transform It!

Time Frame: [90 min].

Lesson Objectives:

By the end of this lesson students will be able to:

- Explain why it's important to throw trash in the right bins.
- Compare between recyclable and non-recyclable items.
- Show how recycling can turn old things into something new.
- Think of smart ways to sort trash better, like using labels or signs.
- Use an AI-powered mapping tool to generate and analyze a world map of recycling bin color codes, comparing similarities and differences across countries.
- Identify and compare physical vs. chemical changes.

Challenge questions of the lesson

1. Why do students avoid using trash bins?
2. How would the school environment change if everyone sorted their waste correctly?
3. If you could design a perfect waste management system for your school, what features would you include?

SEL Relation

- **Community Awareness:** Understand how smart technology, like our Smart Bin, fits into community waste management for a healthier environment.
- **Responsibility:** Recognize our personal role in properly sorting trash, recycling, and contributing to a cleaner school and planet.
- **Collaboration:** Work together during the coding game, discussions, and poster creation to foster a cleaner, more sustainable environment.
- **Appreciation:** Value smart technologies and the resources saved through effective recycling and repurposing.
- **Self-Awareness:** Reflect on learning about smart waste management and identify areas for personal growth in environmental actions.
- **Communication:** Clearly express ideas about smart waste solutions through speaking, drawing, and writing.
- **Critical Thinking & Problem-Solving:** Analyze waste problems, develop solutions through the coding game, and creatively address waste management.
- **Environmental Literacy/Responsibility:** Deepen understanding of waste's environmental impact and take action through smart waste management and recycling.

Engage

Teacher Role:

- **Facilitator:** Guide a discussion about the waste problem introduced in the story and its real-world implications in the school environment.
- **Questioning:** Ask guiding questions to encourage critical thinking about the causes of the waste problem and potential initial steps for investigation.
- **Presenter:** Introduce the concept of identifying and understanding a problem as the first step in a broader problem-solving or design thinking approach.

Expected From Students:

- **Comprehension:** Actively listen to or read the story about the schoolyard waste problem and identify the main characters, the core issue presented, and Mr. Yusef's frustration.
- **Problem Identification:** Recognize the persistent waste issue in the schoolyard and understand why it is a problem that Adam and Laila feel needs addressing.
- **Critical Thinking & Inquiry:** Discuss and reflect on the reasons given for the trash problem's persistence and interpret Adam and Laila's decision to investigate further.
- **Narrative Connection:** Answer questions related to the story to demonstrate understanding of the narrative and the characters' response to the problem.

Parts of the books included.

Adam (frowning): Look at this mess, Laila! The schoolyard is covered in trash.
Laila (sighing): Yeah, no one seems to care.
(They notice Mr. Yusef, the janitor, picking up trash, looking exhausted.)
Mr. Yusef (muttering): I clean every morning, but it never lasts...
Adam (concerned): Mr. Yusef looks upset. Let's talk to him.
(They walk over.)



Adam (shocked): What?! We can't lose break?
Laila (determined): We need to fix this.
Adam: Let's investigate why students don't use the bins.

STEAM SCOUTS

Story Reflection

1. Why was Mr. Yusef frustrated in the schoolyard?
A. Students were playing too loudly.
B. The trash bins were missing.
C. The schoolyard was always messy with trash.
D. The principal asked him to plant more trees.
2. What did Adam and Laila decide to do after talking to Mr. Yusef?
A. Ignore the problem and play.
B. Start investigating why students don't throw trash properly.
C. Tell the principal to close the playground.
D. Ask their parents to clean the schoolyard.

Let's Think!

Uh-oh! There's trash on the floor instead of in the bin. Can you think of three different ways to help your classmates understand why it's important to throw garbage in the right place?

(You can write a short speech, draw a poster, or even make a comic!)



Figure 1: Clean-Up Campaign - Inspiring Students to Take Action.

STEAM SCOUTS

Imagine you're designing a "Sorting Station" for the classroom! What labels, pictures, or colors would you use on each bin to show clearly where each type of waste should go?



Figure 2: Designing a Classroom Sorting Station for Waste Management.

Story



1. Set the Scene with Emotion and Curiosity.

- **Teacher says (with a tone of concern, then curiosity):**
 - "Class, today Adam and Laila have something important to share with us a problem they've noticed right here in their schoolyard. Can you imagine what that might be? What do you usually do when you see something messy or a problem that needs a clever idea to fix?"
 - (Let children respond with ideas about cleaning, organizing, or thinking of new ways to do things.).
- **Then:**
 - "Yes! And today, Adam and Laila haven't just found a problem; they've even started thinking about how to fix it. Let's listen closely to their idea and see if we can help them figure out how to make our schoolyard cleaner..."

2. Engage with the Problem Dialogue.

- **Purpose:** Instead of reading a narrative story, this section uses the provided dialogue as a mini-play or shared reading to introduce the core problem and spark initial ideas for understanding and addressing it.
- **Implementation:**
 - **Read Aloud with Roles:** Assign roles (Adam, Laila, Mr. Yusef) and read the dialogue aloud with expression. Encourage students to use different voices for each character.
 - **Pause and Discuss:** After each character's line (or set of lines), pause and ask guiding questions:
 1. "What problem did Adam notice in the schoolyard?"
 2. "What problem did Laila see Mr. Yusef having?"
 3. "What reasons does Mr. Yusef give for the schoolyard being messy?"
 4. "What do Adam and Laila decide they need to do?"
 5. "What does Adam suggest as a first step?"

Possible Strategies for Implementation:

1. Role-Play the Problem & Solution Brainstorm (Social-Emotional Learning & Problem Identification).

- Have students role-play the provided dialogue. Encourage them to express the feelings of frustration about the mess and the determination to fix the problem.
- Extend the role-play by having them imagine more ideas for how to investigate why students don't use the bins, fostering initial problem-solving brainstorming.

2. Vocabulary and Problem/Solution Language (English Language Arts)

- Highlight and practice key words from the dialogue and the problem: "messy", "frustrated", "upset", "warned", "fix", "investigate".
- Let students form simple sentences to describe the problem: "The schoolyard is messy" or "Mr. Yusef looks exhausted".
- Then, encourage sentences for the solution: "We need to fix this" or "Let's investigate why students don't use the bins".

3. Problem Mapping or Idea Sketching (Critical Thinking & Visual Literacy)

- Provide students with a simple "Problem/Solution" chart or drawing space.
- On one side (Problem), ask them to draw or write about the messy schoolyard.
- On the other side (Solution), ask them to draw or write their initial ideas for how Adam and Laila might investigate the problem or propose initial ways to fix it, based on the dialogue. This helps them visualize the challenge and the proposed solution.

Story Reflection



Objective:

Ensures students comprehend the story and recognize how messy environments impact the feelings of individuals and the importance of finding collective solutions for shared spaces.

Answer:

Story Reflection



1. Why was Mr. Yusef frustrated in the schoolyard?
 - A. Students were playing too loudly.
 - B. The trash bins were missing.
 - C. The schoolyard was always messy with trash.**
 - D. The principal asked him to plant more trees.
2. What did Adam and Laila decide to do after talking to Mr Yusef?
 - A. Ignore the problem and play.
 - B. Start investigating why students don't throw trash properly.**
 - C. Tell the principal to close the playground.
 - D. Ask their parents to clean the schoolyard.

1. Why was Mr. Yusef frustrated in the schoolyard?

- Expected Answer: C) The schoolyard was always messy with trash.
- Reasoning: The story explicitly states, "Adam: The schoolyard is always messy," and "Laila: Students don't sort their trash. Mr. Yusef is frustrated." This directly links Mr. Yusef's frustration to the messiness and unsorted trash in the schoolyard.

2. What did Adam and Laila decide to do after talking to Mr. Yusef?

- Expected Answer: B) Start investigating why students don't throw trash properly.
- Reasoning: After Mrs. Sara asks, "Why do you think this happens?" and students offer reasons like "They don't know where to throw things" and "Bins get too full," Laila suggests "We need a better system!" and Adam says "Maybe a smart bin?" Mrs. Sara concludes with "Great idea! Let's research." This clearly indicates their decision to investigate the problem and potential solutions, rather than immediately cleaning or building.

Possible Strategies for implementation:

1. Think-Pair-Share:

- **Purpose:** To make thinking active and inclusive for all learners, specifically for reflecting on the "Story Reflection" questions.
- **How to implement:**
 - **Think:** Give students time to silently reflect on the "Story Reflection" questions:
 1. "Why was Mr. Yusef frustrated in the schoolyard?"
 2. "What did Adam and Laila decide to do after talking to Mr. Yusef?"
 - **Pair:** Have them discuss their thoughts and answers to the reflection questions with a partner.
 - **Share:** Each pair shares their best answers and insights with the class, explaining their reasoning based on the story.

Let's Think!



Objective:

Students will creatively design solutions for a messy schoolyard, promoting design thinking, problem-solving, and responsible citizenship.

Possible answers:

1. Short Speech:

"Hi friends! Did you know that throwing trash on the ground can hurt animals and make our school look messy? If we all put garbage in the right bins, we help keep our Earth clean and safe. Let's work together to take care of our school!"

2. Poster Idea (with captions):

- Picture of a clean schoolyard with smiling kids
 - Caption: "Clean spaces = happy faces!"
- Picture of trash on the ground and a sad Earth
 - Caption: "Litter hurts our planet."

- Picture of a student throwing trash in a bin
 - Caption: "Be a trash hero!"

3. Comic Strip (3 panels):

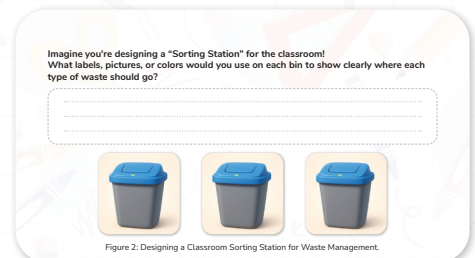
- Panel 1: Jamila walks past overflowing trash, looks upset.
 - Text: "This mess is gross!"
- Panel 2: Adam points to bins and says, "Let's fix this together!"
- Panel 3: Kids sort waste into proper bins.
 - Text: "Yay! Our school is clean again!"



Activity Instructions

Step 1: Set the Stage.

- Say: "Today, we're going to be 'Schoolyard Solution Designers'! We're going to think of ways to make our schoolyard a clean and happy place for everyone."
- Show the provided image of the messy schoolyard (or an actual photo of your schoolyard, if appropriate).

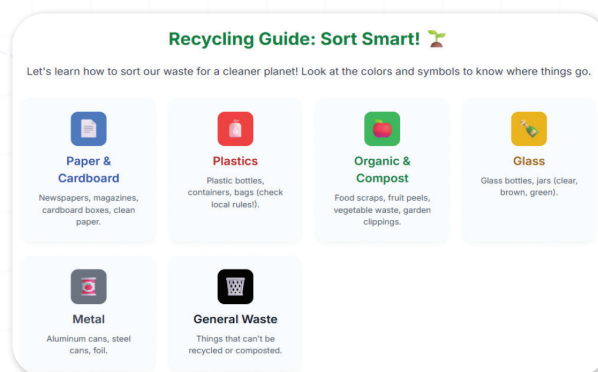


Step 2: Activate Thinking.

- Briefly review the story of Adam and Laila and the problem of the messy schoolyard.
- Ask:
 - "What made the schoolyard messy in the story?"
 - "How did the messy schoolyard make Mr. Yusef and others feel?"
 - "What ideas did Adam and Laila have to help fix the problem?"

Step 3: Explore Materials.

- Distribute:
 - Colours, markers
 - Glue sticks and safety scissors.
 - Labels link to print: <https://gemini.google.com/share/317285ee445a>



You can print the symbols from the previous link and give them to the students to cut out and stick onto the basket pictures in the book.

- **Say:**
 - "You can use labels, symbols, or colours to create your own 'Sorting Station' bins. Think about what would make it really easy for everyone to know where to put each type of trash!"

Step 4: Design Time (Student Work Time).

- **Encourage:**
 - Thinking about clarity and effectiveness (clear labels, universally understood symbols, distinct colours).
 - Creativity in their designs.
 - Collaboration if working in small groups on a shared "Sorting Station" concept.

Step 5: Share & Reflect.

- Let students present their bin designs and explain their reasoning.
- **Prompt with questions:**
 - "What labels, logos, or colours did you choose for your bins, and why?"
 - "How would your 'Sorting Station' help keep our schoolyard clean?"
 - "What's one important message you would tell your classmates about putting trash in the bins?"

Explorer's Toolkit



Teacher Role:

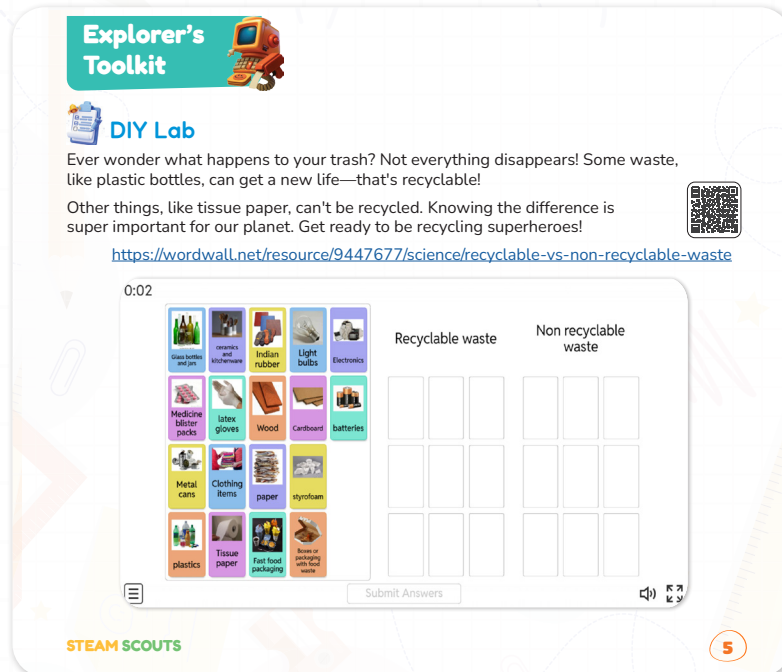
- **Facilitator:** Guide students through the observation process of the messy schoolyard image and encourage them to brainstorm solutions.
- **Questioner:** Ask probing questions to encourage deeper thinking about waste sorting and environmental responsibility ("Why is sorting important?", "How does a clean schoolyard benefit everyone?").
- **Model:** If needed, model how to label one of the bins or explain one reason why putting trash in bins is important.
- **Support:** Offer assistance to students who are struggling with design ideas or articulating their explanations.
- **Connector:** Help students connect their designs and ideas back to the story of Adam and Laila and the bigger picture of keeping their own school environment clean.

Expected From Students:

- **Active Observation:** Students will analyze the messy schoolyard image and understand the problem it presents.
- **Creative Design:** Students will design clear and effective labels, logos, or color schemes for their "Sorting Station" bins.

- **Communication:** Students will articulate (through writing or drawing) reasons why proper waste disposal is important.
- **Problem-Solving:** Students will demonstrate an understanding of how their designs contribute to a solution for a cleaner school environment.

Parts of the book included:



DIY Lab

Objective:

Students will observe and evaluate various waste items to determine if they are recyclable or non-recyclable, developing skills in classification, critical thinking, and environmental responsibility through a hands-on activity.

Materials Needed:

- Access to the interactive Wordwall game: <https://wordwall.net/resource/9447677/science/recyclable-vs-non-recyclable-waste>
- A device (computer, Laptop) for each student or small group to access the game.
- Internet connection.
- Pencils/pens (optional, for notes).



Activity Instructions:

Step 1: Set the Stage - The Mystery of Trash!

- **Say:** "Ever wonder what happens to your trash after you throw it away? Does it just disappear? Today, we're going to become 'Recycling Superheroes' and discover the secret life of trash!"
- **Introduce the concept:** Some waste can get a new life (recyclable!), while other things cannot be recycled. Explain that knowing the difference is super important for our planet.

Step 2: Dive into the DIY Lab - Recyclable vs. Non-Recyclable.

- **Introduce the Wordwall Game:** Explain that you will be using an online game to practice sorting different items.
- **Access the Link:** Guide students to open the provided Wordwall link on their devices.
- **Explore and Sort:** Instruct students to drag and drop each item into the correct category: "Recyclable Waste" or "Non-Recyclable Waste."
- **Encourage Discussion:** As they play, encourage them to think about why certain items go into each category.

Let them fill in the CER chart and answer the questions.



What do you believe is the difference between recyclable and non-recyclable items?

Recyclable items can be broken down and used again to make new products, like paper, glass, and some plastics. Non-recyclable items cannot be reused in this way and usually end up in landfills because they are too difficult or costly to recycle.

What are two examples of items you saw in the game that can be recycled?

1. Plastic water bottles
2. Empty soda cans

How does sorting waste the right way help our planet?

Sorting waste correctly helps reduce pollution, keeps our environment clean, and saves natural resources by reusing materials instead of making new ones from scratch. It also keeps animals safe and helps fight

What are two examples of items that cannot be recycled?

1. Dirty tissue paper
2. Food scraps

Possible Strategies for implementation:

1. Explore with Purpose (Interactive Sorting).

- **Strategy:** "Click, Sort, Explain" .
 - Students actively engage with the Wordwall game, clicking and sorting items.
 - Encourage them to verbally explain their choices as they sort, especially when unsure: "Why do you think this is recyclable?" (Supports classification skills and verbal reasoning).

2. Share What You Learned (Quick Check).

- **Strategy:** "I Discovered That..."
 - After playing the game, ask students to complete the sentence: "I discovered that..." to summarize their key learning about waste.
 - This helps students synthesize their learning into a concise statement.

3. Provide Examples (Game Review).

- **Strategy:** "Waste Watcher List".
 - Have students quickly list or sketch a few items they remember sorting into each category.
 - Provide sentence frames: "From the game, I know that [item name] is recyclable because..." or "[Item name] is not recyclable because [reason]."
 - Encourage the use of specific examples from their game experience.

4. Explain Why It Matters (Planet Protector Chat).

- **Strategy:** "Our Planet's Health" Circle .
 - Sit in a circle (or small groups) and prompt: "Why is it important for us to be good at sorting our trash?"
 - Students use sentence starters: "It's important because..." or "Being a recycling superhero helps our planet by..."

Explain



Teacher Role:

- **Facilitator:** Introduce the video and guide students through the anticipation, focused observation, and reflection activities.
- **Purpose Setter:** Clearly explain the learning objectives and what "superpowers" of trash students should be looking for in the video.
- **Engagement Encourager:** Prompt students to actively observe the recycling process and think critically about the transformation of materials.
- **Discussion Moderator:** Lead the post-watching discussion, ensuring all students have an opportunity to share their observations and insights about recycling.

- **Concept Clarifier:** Address any questions or misunderstandings that may arise about what can or cannot be recycled, or the recycling process itself.
- **Connection Maker:** Help students connect the information in the video to previous activities about sorting waste and the broader importance of environmental responsibility.

Expected From Students:

- **Active Viewing:** Students will watch the video attentively, focusing on identifying different recyclable materials and the steps involved in their transformation.
- **Note-Taking (Optional but encouraged):** Students may jot down key materials, steps in the recycling process, or new "superpowers" of trash they discover.
- **Thoughtful Participation:** Students will contribute meaningfully to the pre- and post-watching discussions, sharing their initial thoughts, observations, and reflections on recycling.
- **Critical Analysis:** Students will think critically about the information presented in the video and how it demonstrates the value of recycling and giving old things a "brand-new life."

Parts of the books included.



Screen Lessons

Do you think trash can have a superpower?
It can, through recycling! Let's discover how we give old things a brand-new life.



https://www.youtube.com/watch?v=6jQ7y_qQYUA&t=1s&ab_channel=KidsAcade



Screen Lessons

Objective:

Students will develop a deeper understanding of the recycling process and its impact by actively engaging with video content through anticipation, focused observation, and structured reflection. They will practice critical thinking, listening, and communication skills as they analyze how materials get a "new life," revise initial assumptions, and articulate new insights about environmental responsibility.

Video link: https://www.youtube.com/watch?v=6jQ7y_qQYUA&t=1s&ab_channel=KidsAcademy



Possible Strategies for implementation:

1. Pre-Watching: Unlock the Superpower!

Strategy: Anticipation Guide - "Trash Superpowers"

- How to Implement:
 - Before showing the video, present students with 2-3 statements related to recycling and its benefits (without revealing if they are true or false). Examples:
 1. "Once we throw something away, it's gone forever."

2. "Recycling helps make new things from old things."

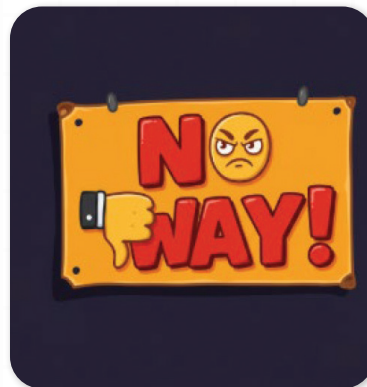
3. "Only plastic can be recycled."

- Ask students to individually agree or disagree with each statement and briefly jot down their reasons before watching the video.
- Explain that the video will help them confirm or revise their initial thoughts about "trash superpowers."

Optional: you can print them these signs to let them more interactive.

<https://g.co/gemini/share/2a43aabb62ab> agree sign.

<https://g.co/gemini/share/c164f1ad7928> disagree sign.



2. During Watching: Observing the Transformation.

Strategy: Focused Observation with Prompts - "Recycling Detective"

- How to Implement:
 1. Before starting the video, provide students with 3-2 specific questions or prompts to focus on while watching. Examples:
 - "What are some different types of materials shown being recycled in the video?"
 - "What steps happen to a material like a plastic bottle after it's put in the recycling bin?"
 - "What new items are made from recycled materials in the video?"
 2. Encourage students to pay close attention to these aspects as they watch. They can take brief notes or draw quick sketches if helpful.

3. Post-Watching: Share Your Insights.

Strategy: Think-Pair-Share (Revisited with Video Focus) - "Recycling Revelation"

- How to Implement:
 - Pose a question related to the video's content ("What was one new 'superpower' of trash you learned about from the video?" or "How did the video change your thinking about throwing things away?").
 - Give students individual "think time" to formulate their answers.
 - Have them pair with a partner to discuss their thoughts and discoveries.

- o Facilitate a whole-class share-out of key insights and "recycling revelations."
- o Facilitate a whole-class share-out of key insights.



Dig Deeper

Objective:

Students will develop a deeper understanding of the recycling process and its impact by actively engaging with an article content through anticipation, focused observation, and structured reflection. They will practice critical thinking, listening, and communication skills as they analyze how materials get a "new life," revise initial assumptions, and articulate new insights about environmental responsibility.

Article link: <https://kids.nationalgeographic.com/science/article/tour-a-recycling-center>

Possible Strategies for implementation:

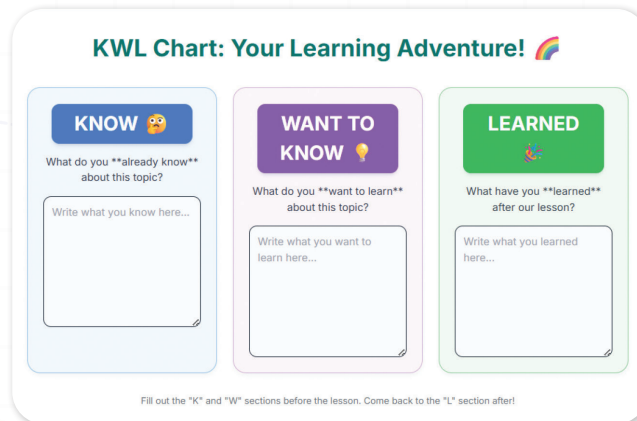


1. Pre-Reading: Activating Prior Knowledge & Setting Purpose

Strategy: KWL Chart - "Recycling Center Expedition"

- How to Implement:
 1. Introduce the article topic: "Today, we're going on a virtual tour of a recycling center through an article! Before we start, let's think about what we already know and what we want to find out."
 2. Distribute or draw a KWL chart on the board (Know, Want to Know, Learned).
 3. "K" (Know): Ask students, "What do you already know about recycling or what happens to our trash after it's picked up?" Have them individually fill in the "K" column with their existing knowledge.
 4. "W" (Want to Know): Next, ask, "What do you want to learn or wonder about a recycling center or how items get recycled?" Have them fill in the "W" column.
 5. Explain that reading the article will help them fill in the "L" column later.

Please: print them the following chart: <https://gemini.google.com/share/9c121e01d974>



KWL Chart: Your Learning Adventure!

KNOW 🧐	WANT TO KNOW 💡	LEARNED 🎉
What do you **already know** about this topic?	What do you **want to learn** about this topic?	What have you **learned** after our lesson?
Write what you know here...	Write what you want to learn here...	Write what you learned here...

Fill out the "K" and "W" sections before the lesson. Come back to the "L" section after!



2. During Reading: Focused Information Gathering

Strategy: Guided Reading with Highlighting/Note-Taking - "Information Detectives"

- How to Implement:
 1. Instruct students to read the article "Tour a Recycling Center | National Geographic Kids" either individually, in pairs, or as a class.
 2. Provide specific prompts to guide their reading. Examples:
 - "As you read, try to find out where the recycling center gets its materials from."
 - "Look for the different steps that happen to recycling once it arrives at the center."
 - "Pay attention to what new products are made from recycled items."
 3. Encourage students to highlight, or jot down brief notes/keywords related to these prompts directly in a paper or their books.

3. Post-Reading: Consolidating Learning & Reflection

Strategy: KWL Chart Completion & Discussion - "Recycling Experts Share"

- How to Implement:
 1. Bring students back together.
 2. "L" (Learned): Ask students to fill in the "L" column of their KWL chart with new information they learned from the article, paying special attention to answering questions from their "W" column.
 3. Share and Discuss: Facilitate a whole-class discussion where students share what they wrote in their "L" column.
 4. Reflect: Ask concluding questions: "What was the most surprising thing you learned about a recycling center?" or "How does knowing what happens at a recycling center make you feel about sorting your trash?"

Elaborate and Evaluate



Teacher Role:

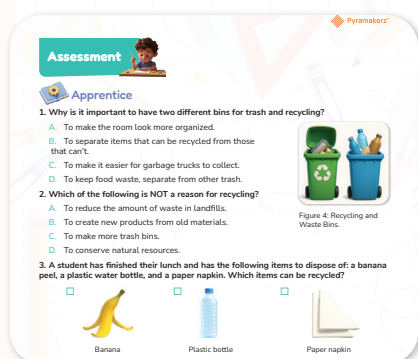
- Reinforce key recycling concepts (recyclable vs. non-recyclable, physical vs. chemical changes).
- Clearly explain the sorting system for the six waste bin colors and connect it to real-world recycling practices.
- Provide instructions and demonstrations for activities (sorting items, building with recycled materials, using AI tools).
- Circulate and monitor students' work, asking guiding questions to check for understanding and push deeper thinking.
- Ensure safety while handling craft or recycled materials (reminding students about masks, gloves, and neat workspace use).
- Encourage creativity and problem-solving by linking recycling knowledge to engineering and AI challenges.

- Assess learning continuously—through observation, questioning, and reviewing student outputs (answers, models, maps).

Expected From Students:

- Apply their knowledge of recycling by sorting items into the correct waste bins.
- Use recycled materials creatively to design and build models (e.g., robots), showing practical understanding of reuse.
- Answer questions and reflect on why recycling is important and how materials are classified.
- Explore deeper ideas such as physical vs. chemical changes in materials.
- Use digital tools/AI responsibly to investigate global waste color codes and generate a world map.
- Collaborate in groups, share findings, and present their creative solutions to peers.
- Show both factual understanding (what belongs where) and higher-level thinking (why global codes differ, how recycling supports sustainability).

Parts of the books included.

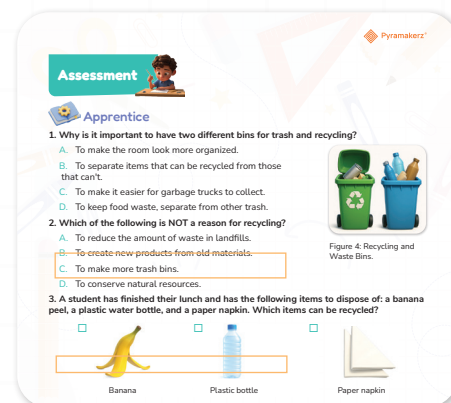


Objective:

Students will analyze questions and images related to trash and recycling to demonstrate their understanding of waste separation, the benefits of recycling, and the correct disposal of common items.

Answers:

1. Why is it important to have two different bins for trash and recycling?
 - B. To separate items that can be recycled from those that can't.
2. Which of the following is NOT a reason for recycling?
 - C. To make more trash bins.
3. A student has finished their lunch and has the following items to dispose of: a banana peel, a plastic water bottle, and a paper napkin. Which items can be recycled?



B. b) plastic water bottle.

Possible Strategies for Implementation:

1. Question Analysis & Discussion:

Strategy: "Think-Pair-Share" for Each Question.

- How to Implement:
 1. Present Question 1: "Why is it important to have two different bins for trash and recycling?"
 2. Think: Give students individual time to read the question and all answer choices, deciding on their best answer privately.
 3. Pair: Have students turn to a partner and discuss their chosen answer and why they think it's correct (or why other options are incorrect).
 4. Share: Facilitate a brief whole-class discussion, asking pairs to share their reasoning before revealing the correct answer. Repeat for Question 2.

2. Visual Identification & Application:

Strategy: "Recycling Detective" with Image Analysis.

- How to Implement:
 1. Present Question 3: "A student has finished their lunch and has the following items to dispose of: a banana peel, a plastic water bottle, and a paper napkin. Which items can be recycled?"
 2. Observe: Direct students to carefully look at each image (banana peel, plastic bottle, paper napkin).
 3. Decide: Ask students to individually check the box below the items they believe can be recycled.
 4. Discuss & Justify: Facilitate a discussion asking students to explain why they checked certain boxes and left others unchecked, drawing on their knowledge from previous activities (videos, articles, sorting games).



Builder

Objective:

- Understand the importance of sorting waste into correct recycling categories.
- Explore the concept of upcycling by transforming waste materials into creative products.
- Apply problem-solving and engineering skills to build a robot model using recycled materials.

Materials Needed:

- Recycled items brought by students (plastic bottles, cardboard, cans, paper, etc.).

- Sorting baskets/bins labeled with color codes:
 - White – Paper.
 - Gray – Glass.
 - Yellow – Electronics.
 - Blue – Metal.
 - Red – Plastic.
 - Orange – Special Waste (purpose left as a challenge).
- Tape, glue, or rubber bands for joining materials.
- Scissors (teacher-supervised).
- Optional: small wheels, bottle caps, or motors to enhance robot design.
- Safety items: gloves, masks.

Activity Instructions:

1. Preparation (before the lesson):

- Ask students in advance to bring safe, clean recyclable items from home for upcycling.
- Arrange sorting baskets labeled with the six recycling color codes.

2. Sorting Challenge:

- Begin the session by showing students the different baskets.
- Guide them to sort their brought materials into the correct bins.
- Pause at the orange basket and tell students: “This one is a mystery today—we’ll investigate its purpose later.”

3. Upcycling Robot Design:

- After sorting, ask students to select some of their materials back (from the safe recyclables).
- Challenge them to upcycle these items into a robot body.
- Encourage creativity: bottles for arms, cardboard for body, caps for wheels, etc.

4. Building & Presentation:

- Students work in groups to build their robot models.
- Once finished, groups present their robots, explaining what materials they reused and how they applied recycling rules.



Mastermind

Objective:

- Investigate global waste color codes and understand similarities and differences across countries.
- Develop digital literacy by using AI-powered tools to research and visualize data.

- Deepen critical thinking by focusing on the mystery of the orange bin (Medical waste).
- Connect recycling knowledge with global awareness and technology use.

Activity Instructions:

1. In-Class Introduction (Teacher):

- Present the challenge: "Now that you know how to sort waste in our classroom, let's see how the rest of the world does it!"
- Show students the six color bins and highlight the orange bin as today's special mystery.
- Explain that they will complete this investigation at home using an AI tool.
- Give a quick demonstration of how to use Gemini (or another AI tool) to ask a question and gather results.

2. At-Home Task (Students):

- Use Gemini (or a similar tool) to research the global waste color codes for paper, glass, electronics, metal, plastic, and special waste.
- Pay close attention to the orange color and try to discover what it represents in different countries.
- Generate or sketch a world map showing waste color codes, labeling how they are used in at least three different regions.

3. Follow-Up (Next Class):

- Students bring their findings (digital map, screenshots, or drawings).
- Share their maps with classmates and compare results.
- Teacher facilitates discussion:
 - "Were the color codes the same everywhere?"
 - "Why do you think some countries use different codes?"
 - "What does this tell us about solving global environmental problems?"



Now I Can

Objective:

Students will reflect on their learning journey in waste management by identifying and expressing what they now understand about proper trash disposal, recycling, and keeping our environment clean. Through self-assessment using a "Now I Can" checklist, students will demonstrate their ability to apply new knowledge and commit to responsible actions.

Activity Instructions

Step 1: Reflect on Our Journey!

- Begin by saying: "You've all become incredible 'Recycling Superheroes' and 'Sorting Stars'! We've learned so much about trash, recycling, and how to keep our school and planet clean. Now, let's think about all the amazing things we 'Now I Can' do and understand!"

- Display the "Now I Can" checklist and read each statement aloud, one item at a time.

Step 2: My Personal Superpowers Checklist.

- Ask students to open their "Now I Can" checklist and provide pencils or colors.
- Ask students to:
 - Check off each statement they feel confident about.
 - Then, choose one of the checked items and either:
- Draw a picture that shows them doing or understanding that thing.
- Write one sentence about what they learned or how they can do it.
 - Say: "Pick one 'superpower' from the list that you feel really good about! (you can tell them to give it a score 1 = Need more practice, 2 = Can do with a little help, 3 = I can do it strongly by myself!) Draw how you use that superpower, or write about what you learned!"

Step 3: Share Our Collective Impact.

- In groups of 3–4, have students share one thing they checked off and explain:
 - "What activity or lesson helped you learn that 'Now I Can' statement?" ("I learned to sort from the DIY Lab game!")
 - "Why do you think that 'Now I Can' superpower is important for our school and the Earth?"
- Facilitate a brief whole-class share-out: Ask a few groups to share one of their "Now I Can" statements and their reflections.

Relation to Standards.

1. NGSS (Next Generation Science Standards).

- **K-LS1-1 (Life Science - From Molecules to Organisms: Structures and Processes).**
 - **Standard:** Use observations to describe patterns of what plants and animals (including humans) need to survive.
 - **Relation:** While not a direct match, the understanding that proper waste management (reducing pollution, keeping environments clean) contributes to healthy ecosystems and human well-being for survival aligns broadly with environmental health.
 - **In this lesson:** Students understand that a clean environment (less trash, more recycling) is better for plants, animals, and people.
- **K-ESS3-3 (Earth and Space Sciences - Earth and Human Activity).**
 - **Standard:** Communicate solutions that will reduce the impact of humans on the land, water, and/or air.
 - **Relation:** This standard is directly aligned with the core focus of the unit. Students learn and communicate solutions for waste reduction and recycling.
 - **In this lesson:** Students design sorting bins, sort waste, explain the importance of recycling, and create posters communicating solutions to reduce human impact (trash) on the environment.

- **K-2-ETS1-1 (Engineering Design - Defining and Delimiting Engineering Problems).**
 - **Standard:** Ask questions, make observations, and gather information about a situation people want to change to define a simple problem.
 - **Relation:** The initial story of the messy schoolyard explicitly sets up a problem-defining scenario.
 - **In this lesson:** Students identify the "messy schoolyard" as a problem, ask "why does this happen?", and observe unsorted trash to define the need for a "better system."
- **K-2-ETS1-2 (Engineering Design - Developing Possible Solutions).**
 - **Standard:** Develop a simple sketch, drawing, or physical model to illustrate how the shape of an object helps it function.
 - **Relation:** The activity where students design "Smart Bins" directly involves developing solutions through drawings or models.
 - **In this lesson:** Students design visual solutions for sorting bins and illustrate how labels/colors function to improve waste separation.
- **K-2-ETS1-3 (Engineering Design - Optimizing the Design Solution).**
 - **Standard:** Analyze data from tests of two objects designed to solve the same problem to compare the strengths and weaknesses of how each performs.
 - **Relation:** While not testing two physical objects, the comparison between current messy systems and proposed "smart bins" or improved sorting methods reflects analyzing potential solutions. The Wordwall game also involves 'testing' their sorting knowledge.
 - **In this lesson:** Students implicitly compare the "messy schoolyard" (current system) with ideas for a "smart bin" or better sorting (improved system) and reflect on how effective different sorting methods (e.g., visual cues) perform.

2. ISTE Standards for Students.

- **Empowered Learner (1a, 1c).**
 - **Standard:** Students articulate and set personal learning goals, develop strategies to achieve them and reflect on the learning process. Students use technology to seek feedback that informs and improves their practice and to demonstrate their learning in a variety of ways.
 - **Relation:** Students actively engage with digital tools (Wordwall game, video), reflect on their learning via "Now I Can," and demonstrate understanding through various outputs (designs, posters, answers).
 - **In this lesson:** Students utilize digital games and videos to learn about recycling, use a checklist to reflect on their learning goals, and demonstrate understanding through their designs and explanations.
- **Innovative Designer (4a, 4b, 4c).**
 - **Standard:** Students know and use a deliberate design process for generating ideas, testing theories, creating innovative artifacts or solving authentic problems. Students select and use digital tools to plan and manage a design process that

considers design constraints and anticipated effects. Students develop, test and refine prototypes as part of a cyclical design process.

- Relation: The entire unit involves identifying a problem (messy schoolyard), brainstorming solutions (smart bins, sorting), designing solutions (bin labels, posters), and refining their understanding.
- In this lesson: Students identify the problem of unsorted trash, design a "Sorting Station," and create educational posters, embodying aspects of the design-solve-communicate cycle.